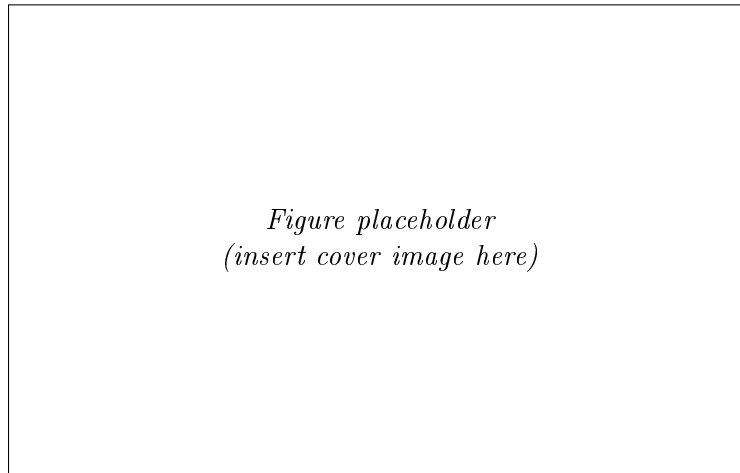


Technical Design Document

Cubli



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1 Introduction

1.1 Background & Problem Statement

The Cubli is a reaction-wheel inverted pendulum: a cube-shaped platform that balances on an edge or a corner using only internal actuation, with no external supports or contact forces beyond the single line or point of contact with the ground. The concept was introduced by the Institute for Dynamic Systems and Control (IDSC) at ETH Zurich [1, 2], whose original prototype ($15 \times 15 \times 15$ cm) mounts one brushless DC motor and flywheel on each of three orthogonal faces. Commanding a change in wheel speed produces, by conservation of angular momentum, an equal and opposite reaction torque on the cube body; braking a wheel abruptly transfers its stored angular momentum to the body in a near-impulsive event, which is what allows the cube to *jump up* from resting on a face into a balance on an edge or corner.

Dynamically, the system is an *underactuated*, nonlinear, unstable-equilibrium control problem: three actuators (the wheels) must stabilize a body with more degrees of freedom than actuated inputs, about equilibria (edge and corner balance) that are open-loop unstable and increasingly coupled across axes as the contact geometry shrinks from a line (edge) to a point (corner). This is structurally the same problem as the classical inverted pendulum, extended from a single rotational axis to a fully three-dimensional, momentum-exchange-actuated body — which is why the Cubli has become a standard benchmark platform in the controls community over the past decade.

The problem this project addresses is therefore twofold: (i) derive and validate a dynamic model and controller capable of stabilizing this underactuated system about its unstable equilibria and driving it through commanded reorientations, and (ii) realize this in hardware as a modular, largely parametric design, since final actuator and sensor specifications are not yet confirmed at the time of writing. The target capability set, in order of increasing difficulty, is edge balance, corner balance, controlled rotation, and walking (Section 1.3).

1.2 Purpose, Application & Mission Context

The Cubli is not presented here as a balancing novelty but as a terrestrial, gravity-loaded demonstrator of **momentum-exchange attitude control** — the same physical mechanism that governs how real spacecraft point themselves. Nearly every three-axis-stabilized satellite holds and changes its attitude using reaction wheels: flywheels whose commanded speed change produces a reaction torque on the spacecraft body, exactly as in the Cubli. Spacecraft typically carry three or four such wheels (often in a pyramidal arrangement for redundancy) — directly analogous to the Cubli’s three orthogonal wheels — and this reaction-wheel approach is the standard for CubeSats and small satellites in particular, making the Cubli representative of a real, currently-flown hardware class rather than an idealized toy problem. Reaction wheels are distinct from control moment gyros (CMGs), which gimbal a constantly-spinning wheel instead of changing its speed to produce much larger torques [3]; positioning the Cubli explicitly as a *reaction-wheel*, not CMG, demonstrator keeps the application claim precise.

Framed as an ADCS problem, the Cubli’s core behaviors map directly onto a spacecraft’s day-to-day pointing task:

- **Edge / corner balance** → regulation to a commanded reference attitude and disturbance rejection — holding a pointing setpoint against perturbations.
- **Controlled rotation** → a slew maneuver: reference tracking to a new commanded attitude, exactly as a satellite re-aims a payload or antenna.
- **Jump / walk** → momentum-exchange hopping mobility, flight-proven on small bodies

where wheeled traction is impossible. JAXA’s Hayabusa2 mission deployed the MINERVA-II1 rovers [4] and the MASCOT lander [5] on asteroid Ryugu (2018), both of which move by spinning up and braking an internal flywheel or eccentric mass to hop across the surface, with MASCOT additionally self-righting the same way. Walking, as a sequence of controlled, directional hops, is a genuine step beyond a single ballistic hop and is an active research frontier (e.g. ETH’s SpaceHopper [6]).

This gives the project two complementary application narratives rather than one: it is simultaneously a low-cost, hardware-in-the-loop testbed for standard spacecraft *attitude determination and control* (ADCS), and a ground analog for the momentum-exchange *surface mobility* now used on small-body missions. Framed against a companion star-tracker project, which answers “*where am I pointing?*” (attitude determination), the Cubli answers “*drive me to that reference and hold it*” (attitude control) — the two halves of a complete ADCS loop.

1.3 Objectives

The project targets four behaviors, deliberately ordered as a difficulty ramp in which each milestone is a control precondition for the next:

1. **Edge balance** — stabilize the cube on a single edge, the least unstable equilibrium (contact is a line, one primary axis of instability).
2. **Corner balance** — stabilize the cube on a single vertex, the fully unstable equilibrium (point contact, all three axes coupled and unstable simultaneously).
3. **Controlled rotation** — execute a commanded slew between equilibria (e.g. edge to edge, or in-place reorientation), demonstrating reference tracking rather than pure regulation.
4. **Walking** — chain repeated, directional controlled falls between edges/corners into a net translation across a surface, re-stabilizing after each transition.

These map one-to-one onto the applications of Section 1.2: (1)–(2) demonstrate pointing regulation and disturbance rejection of increasing difficulty; (3) demonstrates slew/reference-tracking; (4) demonstrates momentum-exchange mobility of the kind flown on Hayabusa2. Milestones (1)–(3) are treated as committed deliverables for this design; walking (4) is scoped as the stretch objective, contingent on hardware confirmation and the jump-up feasibility check (angular momentum required to tip the cube vs. available motor/wheel torque), to be resolved once final components are known.

2 Project Organization

2.1 Team Organization

The project is developed by a multidisciplinary team, with responsibilities distributed across the main technical domains of the Cubli: mathematical modeling, control, mechanical/CAD design, and system integration. Table 1 summarizes each member’s primary role, and the detailed responsibilities are outlined below.

Table 1: Team members, contact details and primary roles.

Member	Contact	Primary Role
Pablo Urioste Alarcón	pb.urioste4@gmail.com +34 654 491 174	System integration and lead: digital actuator and sensor coding, embedded software, and integration of the overall design.
Deyan Nikolaev Vlaev	vlaev.dido@gmail.com +359 877 082 499	Mathematical demonstration and derivation of the system dynamics.
Niccolo	—	Control system design and analysis.
Andrea Liang	andrea.liang.2006@gmail.com +31 06 5796 2976	Support on the control problem and structural/mechanical integration.
Suvanna Viriyanti Wu	Viriyanti.wu@gmail.com +39 329 074 7550	CAD modeling and mechanical design.
Athanasia Nikolova	—	CAD modeling, prototype building/assembly, and product presentation (commercial pitch).

Responsibilities.

- **Pablo Urioste Alarcón** — System integration and overall coordination. Handles the coding of the digital actuators and sensors, the embedded software stack, and the integration of the mechanical, electronic and control subsystems into a working prototype.
- **Deyan Nikolaev Vlaev** — Mathematical foundation. Responsible for the analytical demonstration and derivation of the equations of motion and the dynamic model of the Cubli.
- **Niccolo** — Control. Leads the design, tuning and analysis of the control strategy used to balance and maneuver the Cubli.
- **Andrea Liang** — Control and mechanics support. Assists on the control problem and contributes to the structural and mechanical integration of the system.
- **Suvanna Viriyanti Wu** — CAD and mechanical design. Develops the 3D models and mechanical drawings of the structure and reaction-wheel assembly.
- **Athanasia Nikolova** — CAD, prototype building and commercialization. Contributes to the 3D modeling, leads the physical building and assembly of the prototype, and is responsible for presenting and pitching the product.

2.2 Timeline and Contingency Plan

3 System Analysis & Requirements

3.1 Scope of Work & Use Cases

3.2 Actuation Trade-off — Why Reaction Wheels

3.3 Requirements & Objectives Traceability

Each project objective (Section 1.3) imposes a required capability, which in turn is confirmed by a specific test in the validation plan (Section 6). The mapping below keeps every requirement traceable from motivation to verification.

Table 2: Objective-to-requirement-to-verification traceability.

Objective	Required capability	Verified by
Edge balance	Regulation about a line-contact equilibrium	Sec. 6
Corner balance	Regulation about a point-contact equilibrium, all axes	Sec. 6
Controlled rotation	Reference tracking / slew between equilibria	Sec. 6
Walking (stretch)	Chained jump-up + re-stabilization	Sec. 6

3.4 Coordinate Frames & Conventions

4 Dynamic Modeling & Control

4.1 Modeling Approach: Why 2D First

The full Cubli is an underactuated, nonlinear system that is unstable about all three axes at once when balanced on a corner (Section 1.3). Rather than commit directly to the fully coupled three-dimensional cube, this project first develops and validates the complete modeling, identification, and control workflow on a two-dimensional (2D) single-face prototype. The 2D rig reduces the problem to a single rotational degree of freedom in a vertical plane, which de-risks three things at the scale of one face before the full cube is built: (i) the structure of the equations of motion and the control law, (ii) the parameter-identification procedure that reconciles CAD-predicted properties with the physical frame, and (iii) the jump-up mechanism. The dynamics validated on the 2D model then generalize, axis by axis, to the 3D cube in Section 4.6.

4.2 2D Reaction-Wheel Pendulum

A two-dimensional (2D) prototype will be built at the scale of a single face of the cube, in order to develop and validate the control algorithms before the full cube is assembled. The prototype will consist of a 3D-printed plate matching the dimensions of one cube face, with a momentum wheel and motor mounted at its center and a braking mechanism at one corner. The plate will rest on a bearing at that corner, giving it a single rotational degree of freedom within a vertical plane.

4.2.1 Equations of Motion

The nonlinear equations of motion governing the system will be

$$\ddot{\theta}_b = \frac{(m_b l_b + m_w l) g \sin \theta_b - T_m - C_b \dot{\theta}_b + C_w \dot{\theta}_w}{I_b + m_w l^2}, \quad (1)$$

$$\begin{aligned} \ddot{\theta}_w = & \frac{(I_b + I_w + m_w l^2)(T_m - C_w \dot{\theta}_w)}{I_w(I_b + m_w l^2)} \\ & - \frac{(m_b l_b + m_w l) g \sin \theta_b - C_b \dot{\theta}_b}{I_b + m_w l^2}, \end{aligned} \quad (2)$$

with the motor current-torque relation $T_m = K_m u$, where θ_b is the tilt angle of the pendulum body, θ_w the rotational displacement of the momentum wheel relative to the body, m_b and m_w the masses of the body and wheel, I_b and I_w their moments of inertia about the pivot and motor axis respectively, l the distance from the motor axis to the pivot, l_b the distance from the body's center of mass to the pivot, g the gravitational acceleration, T_m the motor torque, C_b and C_w the dynamic friction coefficients of the body and wheel, K_m the motor torque constant of the selected motor, and u the motor current input commanded through the motor driver.

4.2.2 Equilibria & Linearization

4.3 Parameter Identification & the 2D Testbed

Since the pendulum body and wheel will be 3D-printed, their geometry will be fully defined in CAD prior to fabrication. This will allow m_b , m_w , I_b , I_w , and l_b to be computed directly from the CAD model's mass-properties simulation, given the chosen print material, infill density, and the mass and placement of mounted hardware such as the motor driver, encoder, microcontroller, IMU, and wiring. These simulated values will be cross-checked experimentally once the prototype is fabricated: l_b will be verified by freely hanging the body from different corners, and I_b will be verified by fixing the wheel, hanging the assembly upside down, letting it swing, and fitting the identified model $(I_b + I_w + m_w l^2)\ddot{\theta}_b = -C_b \dot{\theta}_b + (m_b l_b + m_w l)g \sin \theta_b$ to the logged tilt data.

The damping coefficients C_b and C_w cannot be obtained from simulation, since they depend on bearing friction, wiring drag, and motor cogging not captured in CAD, and will therefore be identified purely experimentally: C_w (together with I_w , as a check against the simulated value) by driving the momentum wheel with current steps while the body is clamped and fitting $I_w\ddot{\theta}_w = K_mu - C_w\dot{\theta}_w$ to the logged wheel speed; C_b from the swing-test data described above.

Table 3: Prototype parameters and determination method.

Symbol	Description	Determination method
l	Motor axis to pivot distance	Design choice
l_b	Body CoM to pivot distance	Design choice; verified by hang test
m_b	Pendulum body mass	CAD mass-properties
m_w	Momentum wheel mass	CAD mass-properties
I_b	Body moment of inertia	CAD mass-properties; verified by swing test
I_w	Wheel moment of inertia	CAD mass-properties; verified by current-step fit
C_b	Body damping coefficient	Experimental (swing-test fit)
C_w	Wheel damping coefficient	Experimental (current-step fit)
K_m	Motor torque constant	Motor datasheet

4.4 Jump-Up Feasibility Analysis

With I_b , I_w , m_b , l_b , m_w , and l established from CAD simulation, the wheel angular velocity ω_w required for the jump-up maneuver will be estimated before fabrication. Assuming a perfectly inelastic collision between wheel and body, conservation of angular momentum during impact together with conservation of energy from the moment of impact to the edge-standing position gives

$$\omega_w^2 = (2 - \sqrt{2}) \frac{I_w + I_b + m_w l^2}{I_w^2} (m_b l_b + m_w l) g. \quad (3)$$

This simulation-derived value of ω_w will be used to verify, ahead of fabrication, that the selected motor can reach the required speed within the available bus voltage, and to size the wheel’s inertia — via its radius and any added rim mass — so that the required speed remains within the motor’s practical operating range without excessively reducing the recovery angle available for balancing. Once the prototype is built, the current-step and swing-test identification described above will be used to refine ω_w experimentally, together with an iterative trial-and-error adjustment to account for any deviation from the idealized inelastic-collision assumption (e.g., a non-zero coefficient of restitution or parameter uncertainty).

The braking mechanism will consist of a servo-driven barrier that is swung into a bolt head attached to the momentum wheel to stop it abruptly. The servo’s response time will be a design constraint on the validity of the “instantaneous stop” assumption underlying the jump-up equation above; a slower stop than assumed will yield a lower effective post-impact body velocity than predicted, which will be compensated for through the same iterative tuning process rather than relying on the analytical estimate alone.

4.4.1 Torque-Limited Wheel-Inertia Target

The idealized relation above returns the required wheel speed for a given inertia, but it assumes an instantaneous, lossless momentum transfer. For preliminary sizing we invert the problem—given the achievable motor speed, *what per-wheel inertia does the maneuver demand?*—using a momentum budget that additionally penalizes the finite brake torque. The maneuver must supply enough post-impact angular momentum to raise the cube’s center of mass from its resting

contact to the balancing equilibrium, scaled by an energy margin η . This defines an ideal per-wheel angular momentum

$$h_{w,\text{ideal}} = \sqrt{\eta} \frac{\sqrt{2g\lambda\kappa}}{n} ML^{3/2}, \quad (4)$$

where M and L are the cube mass and edge length, g is gravity, and κ , λ , n are dimensionless geometric factors set by the contact case: the normalized moment of inertia about the pivot (κ), the relative rise of the center of mass to the equilibrium (λ), and the momentum-transfer factor of the contact (n). The corner case ($\kappa = \frac{11}{12}$, $\lambda = \frac{\sqrt{3}}{2} - \frac{1}{2}$, $n = \sqrt{2}$) governs the design as it is the more demanding of the two; the edge case ($\kappa = \frac{2}{3}$, $\lambda = \frac{1}{\sqrt{2}} - \frac{1}{2}$, $n = 1$) is less strenuous.

A brake of finite torque τ_b cannot deliver the impulse instantaneously: during the stop the cube must still overcome the gravitational tip-over torque $\tau_g = MgL/2$, which sets a hard floor $\tau_b > \tau_g$ below which the cube can never tip. The resulting loss is captured by a brake-amplification factor

$$\beta = \frac{\tau_b}{\tau_b - \tau_g}, \quad h_w = \beta h_{w,\text{ideal}}, \quad I_{w,\text{target}} = \frac{h_w}{\omega_{\text{max}}}, \quad (5)$$

where ω_{max} is the maximum practical wheel speed. Table 4 lists the resulting target for the current baseline cube. This per-wheel inertia $I_{w,\text{target}}$ is the requirement that the reaction-wheel design of Section 5.3.3 must meet.

Table 4: Torque-limited wheel-inertia target (corner case, baseline cube).

Symbol	Description	Value
M	Cube mass	1.45 kg
L	Cube edge length	180 mm
η	Energy margin	1.05
τ_b	Brake torque per wheel	5.0 N m
τ_g	Gravitational tip-over floor	1.28 N m
β	Brake-amplification factor	1.34
ω_{max}	Max wheel speed	8400 rpm (880 rad/s)
h_w	Required angular momentum	0.277 kg m ² /s
$I_{w,\text{target}}$	Target wheel inertia	3.15×10^{-4} kg m²

4.5 Control Design on the 2D Model

4.5.1 LQR Baseline Control

4.5.2 Gain Scheduling & Mode Transitions

4.5.3 Jump-Up Strategy

4.6 Generalization to the 3D Cube

The dynamics, identification workflow, and control approach validated on the 2D model extend to the full three-dimensional cube, where three orthogonal reaction wheels must stabilize the body about its edge and corner equilibria.

4.6.1 Full-Body Equations of Motion

4.6.2 Axis Coupling

4.6.3 Edge vs. Corner Equilibria & Linearization

5 Preliminary Hardware Design

This design is preliminary: final actuator and sensor specifications are not yet confirmed, so the mechanical and electrical architecture is kept modular and largely parametric (Section 1.1).

5.1 System Architecture

5.2 Mass and Inertia Budget

Before committing to component geometry, the preliminary design is bounded by three coupled constraints that must be satisfied simultaneously. Sizing to the worst case of each fixes the envelope within which the mechanical and electrical design then iterate.

Volume constraint. Every subsystem—the three orthogonal reaction-wheel/motor modules, the battery, and the electronics stack—must fit inside the 180 mm cube envelope. The binding dimension is the reaction wheel: its outer diameter must clear the internal frame and corner/edge contact features while leaving room for the motor and the servo brake. This caps the usable wheel diameter and, with it, the radius at which ballast mass can be placed—directly limiting the inertia achievable at a given wheel mass (Section 5.3.3).

Mass constraint. The cube is designed to a total mass target of $M = 1.45$ kg. This is not merely a logistics figure: it feeds back into the dynamics, since the gravitational tip-over torque $\tau_g = MgL/2$ and the required jump-up angular momentum both scale with M (Section 4.4.1). A heavier cube demands a more capable brake and a larger wheel inertia, so the mass target and the inertia target are coupled and must be closed together. The preliminary allocation is given in Table 5, with the reaction-wheel set (the one firmly-sized item) committed and the remaining subsystems held as allowances to be refined as components are selected.

Table 5: Preliminary mass budget (per cube). Reaction wheels sized; others allocated.

Subsystem	Allocated mass (g)	Status
Reaction wheels (3×)	395	Sized (Sec. 5.3.3)
Motors (3×)	TBD	Allowance — pending motor selection
Frame / structure	TBD	Allowance — pending CAD
Battery	TBD	Allowance — pending capacity sizing
Electronics (MCU, drivers, IMU, servo)	TBD	Allowance
Wiring / fasteners / margin	TBD	Allowance
Total (target)	1450	Design target

Inertia constraint (corner-critical). Each reaction wheel must store enough angular momentum at the motor’s maximum speed to execute the jump-up maneuver. The two contact cases are not equally demanding: the corner equilibrium requires a larger rise of the center of mass and a less favorable momentum-transfer geometry than the edge equilibrium, so for a common maximum wheel speed $\omega_{\max}$ it demands the greater wheel inertia. Evaluating the torque-limited budget of Section 4.4.1 for both cases (Table 6), the corner case requires 3.15×10^{-4} kg m² against 2.85×10^{-4} kg m² for the edge —about 10% more. The corner jump-up is therefore the critical condition, and the reaction wheel is sized to it; a wheel that satisfies the corner requirement satisfies the edge requirement with margin.

Table 6: Required per-wheel inertia by contact case, at $\omega_{\max} = 8400$ rpm.

Contact case	h_w (kg m ² /s)	$I_{w,\text{target}}$ (10 ⁻⁴ kg m ²)
Edge	0.251	2.85
Corner (critical)	0.277	3.15

5.3 Mechanical Design

5.3.1 2D Prototype Build

5.3.2 3D Cube Frame

5.3.3 Reaction-Wheel Sizing

The reaction wheel must supply the per-wheel inertia $I_{w,\text{target}} = 3.15 \times 10^{-4}$ kg m² derived from the jump-up budget of Section 4.4.1, while remaining light enough not to dominate the cube mass. Rather than adjusting inertia by scaling a solid disc, the wheel is built as a printed *ring plus spokes* whose inertia is then tuned by *ballast*: standard steel hex nuts dropped into hexagonal pockets distributed around the ring. Each pocket both removes plastic and adds steel, so the net inertia gain per pocket is large and the design target is reached by choosing how many nuts to install—a discrete, easily-adjustable knob that also lets the physical wheel be re-tuned after fabrication to absorb parameter uncertainty.

The structure is printed in PET-CF ($\rho = 1290$ kg/m³): a 120 mm outer-diameter ring of 15 mm radial width and 10 mm thickness, carried by three radial spokes 10 mm wide of the same thickness. The hub and motor interface are neglected in the inertia budget. The ballast is standard M5 steel hex nuts ($\rho = 7850$ kg/m³, 11 mm across flats, measured mass 3.12 g), seated in blind hexagonal pockets machined at the mid-width of the ring. Accounting for the plastic each pocket displaces, the *net* contribution per installed nut is +2.39 g of mass and $+1.55 \times 10^{-5}$ kg m² of inertia at the ring radius.

Table 7 sweeps the outer diameter, solving for the number of nuts needed to reach $I_{w,\text{target}}$ and checking that the pockets physically fit. Small wheels need so many nuts that no plastic wall survives between adjacent pockets; wheels of 140 mm and above reach the target from the bare ring alone (no tuning margin, and heavier). The selected design is **OD = 120 mm with 21 nuts**: it meets the inertia target with a ~ 2.5 mm circumferential wall between pockets, splits the inertia roughly evenly between structure (45%) and ballast (55%) so the wheel can be trimmed in either direction, and keeps the three-wheel set to ~ 395 g (27% of the 1.45 kg cube).

Table 7: Reaction-wheel outer-diameter trade study. Selected design in bold.

OD (mm)	Nuts N	Wheel mass (g)	Wheel I_w (10 ⁻⁴ kg m ²)	I_w/target (%)	3 wheels (g)	Pocket fit
80	105	300.6	3.17	100.7	901.7	fails — no wall
90	72	229.6	3.16	100.5	688.7	fails — no wall
100	51	187.3	3.25	103.3	561.9	fails — no wall
110	33	152.2	3.22	102.3	456.7	fails — thin wall
120	21	131.5	3.31	105.2	394.6	OK (~ 2.5 mm)
130	9	110.8	3.23	102.7	332.4	OK (loose)
140	0	—	3.23	—	—	bare ring exceeds target
150	0	—	4.07	—	—	bare ring exceeds target

Radial fit caveat. The circumferential fit is satisfied at 120 mm, but the *radial* fit is marginal: an 11 mm across-flats pocket in a 15 mm-wide ring leaves only 1.8 mm of wall on each side

(dropping to 0.9 mm if a nut is seated corner-out), below the 2 mm minimum-wall target. The ring width should therefore be increased to $\gtrsim 17$ mm, or the nut orientation constrained flats-radial, in the next design iteration. This does not change the inertia budget, only the ring's structural robustness around the pockets.

5.4 Electrical Design

5.4.1 Components

The electronics and software will be built around a microcontroller communicating over a CAN bus with a motor driver that closes the current loop for the selected brushless motor. Wheel speed will be measured with a magnetic encoder on the motor shaft, and body tilt will be estimated from an onboard IMU. Braking will be actuated by a digital servo driven directly from the microcontroller, and a wireless module will provide a telemetry and debugging link. Power will be supplied by a LiPo battery, stepped down through a DC-DC converter for the logic-level electronics, with the motor driver powered directly from the battery bus.

5.4.2 Electrical Diagram

5.5 System Integration

6 Testing and Validation Plan

This plan covers system-level validation. Parameter-identification tests that feed the model directly (swing test, current-step identification) are described with the model they validate in Section [4.3](#).

6.1 Simulation-First Validation

6.2 Bring-up Sequence

6.3 Sensor Testing

6.4 Milestone Acceptance Criteria

6.5 Control Performance Evaluation

7 Future Work

7.1 NMPC & Warm-Start Control (Future Implementation)

7.2 Autonomous Stand-up & Disturbance Rejection

7.3 Single-Wheel Variant / Extensions

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